

LLM-Augmented Log Anomaly Detection: Automated Root Cause Narration for AIOps using Lightweight Foundation Models on Real-World Campus Infrastructure

Trinh Hoang Tu¹ and Cao Tien Thanh^{1*}

Faculty of Information Technology, Ho Chi Minh City University of Foreign
Languages - Information Technology (HUFLIT), Ho Chi Minh City, Vietnam
{tth.csec2005@gmail.com, thanhct@huflit.edu.vn}

Abstract. Modern enterprise campus networks generate massive volumes of heterogeneous, unstructured log streams across multi-site infrastructures. While deep learning models achieve high accuracy in sequence anomaly detection, traditional AIOps platforms suffer from a critical explainability bottleneck: raw reconstruction error scores fail to communicate actionable root causes, forcing Security Operations Center (SOC) analysts to manually triage thousands of log messages under severe time constraints. In this paper, we propose a novel **LLM-Augmented AIOps Framework** synthesizing streaming Drain3 log template parsing, a hybrid Temporal Convolutional Network and Transformer (TCN-Transformer) autoencoder, and a lightweight quantized Foundation Model (Phi-3 / Mistral via INT4 LoRA fine-tuning) for real-time automated Root Cause Analysis (RCA) narration. Evaluated on a production 5GB enterprise campus log dataset from HUFLIT alongside standard benchmarks (BGL, HDFS), our framework achieves a detection F1-score of 0.952, reduces average incident triage time by > 88% (1.15 s LLM narration latency vs 15.4 minutes human manual inspection), and delivers high-fidelity ROUGE-L score (0.842) compared to expert ground-truth playbooks. All source code and benchmark evaluation scripts are publicly released as an open research artifact.

Keywords: AIOps · Log Anomaly Detection · Foundation Models · Root Cause Analysis · TCN-Transformer · Drain3 · Enterprise Campus Infrastructure.

1 Introduction

Higher education and enterprise ICT infrastructures produce high-density syslog telemetry across firewalls, active directory domain controllers, DNS servers, and core routing backbones. Rapid anomaly identification is essential for preventing lateral threat movement, credential theft, and network outages.

* Corresponding author.

Recent advancements in deep sequence modeling—such as DeepLog, LogAnomaly, and Transformer Autoencoders—have dramatically improved anomaly detection accuracy. However, existing approaches present a significant operational bottleneck:

1. **Explainability Deficit:** Anomaly detectors output numeric reconstruction errors or probabilities (e.g., $RE = 0.942$), which fail to explain *why* an anomaly occurred or *what* specific attack vector is active.
2. **Prolonged Human Triage:** SOC analysts must manually search raw log repositories, correlate multi-service timestamps, and draft mitigation steps, leading to Mean Time to Respond (MTTR) exceeding 15 minutes.

To bridge this gap, we present an **LLM-Augmented AIOps Framework** engineered specifically for enterprise campus environments. By coupling Drain3 streaming template extraction and TCN-Transformer sequence modeling with a lightweight INT4-quantized Large Language Model (LLM), our architecture automatically generates natural language Root Cause Analysis (RCA) narratives and structured mitigation playbooks within milliseconds of anomaly detection.

The key contributions of this paper include:

- **End-to-End AIOps Architecture:** A multi-stage pipeline combining Drain3 template extraction, TCN-Transformer sequence anomaly detection, and RAG-guided LLM narration.
- **Lightweight Foundation Model RCA:** Fine-tuning quantized Phi-3 / Mistral models to produce structured JSON incident summaries, attack vector explanations, and remediation steps.
- **Evaluation on 5GB Real-World Production Campus Logs:** Rigorous empirical benchmarking using a 5GB enterprise campus network log dataset collected from HUFLIT alongside BGL and HDFS benchmarks.

2 System Architecture & Methodology

The overall workflow of the proposed system is depicted in Fig. 1.

2.1 Streaming Drain3 Log Parsing

Raw log messages $L = \{l_1, l_2, \dots, l_n\}$ are ingested in real-time. Drain3 uses a fixed-depth parse tree to sanitize variable tokens (IP addresses, timestamps, numerical IDs) into structured template identifiers $T_i \in \mathcal{V}_{\text{vocab}}$.

2.2 TCN-Transformer Sequence Autoencoder

Parsed template sequences $S = [T_1, T_2, \dots, T_k]$ are embedded into vector representations $E \in \mathbb{R}^{k \times d}$. 1D dilated causal convolutions extract short-term temporal dependencies:

$$H_{\text{TCN}} = \text{ReLU}(\text{Conv1D}_{\text{dilated}}(E)) \quad (1)$$

End-to-End LLM-Augmented AIOps Pipeline Architecture

Fig. 1. End-to-End LLM-Augmented AIOps Pipeline: Streaming Drain3 Parsing, TCN-Transformer Anomaly Detection, and Foundation Model RCA Narration.

A multi-head Transformer Encoder processes H_{TCN} to capture long-range contextual relationships. Reconstruction loss over template vocabulary $\mathcal{V}$ yields the sequence anomaly score S_{anomaly} :

$$S_{\text{anomaly}} = \frac{1}{k} \sum_{i=1}^k \mathcal{L}_{\text{CE}}(\hat{y}_i, y_i) \quad (2)$$

2.3 Foundation Model RCA Narration Engine

When $S_{\text{anomaly}} > \tau$, the anomalous log window, template context, and metadata are formatted into a structured prompt $\mathcal{P}$. A quantized Phi-3 / Mistral LLM generates a JSON response containing:

1. **Severity Level:** {CRITICAL, HIGH, MEDIUM, LOW}
2. **Root Cause Summary:** Concise description of the underlying fault or attack.
3. **Remediation Playbook:** Ordered list of immediate counter-measures.

3 Experimental Evaluation

3.1 Dataset Specification

We evaluated our framework on three distinct datasets detailed in Table 1.

3.2 Detection Accuracy Comparison

Table 2 benchmarks TCN-Transformer against baseline log detection models. Our hybrid architecture achieves optimal F1-score (0.9520) with low inference latency (0.0125 ms per sequence).

Table 1. Benchmark Dataset Statistics

Dataset	Raw File Size	Total Log Lines	Environment
HUFLIT Campus Logs	5.12 GB	14,850,220	Real-World Enterprise
BGL (BlueGene/L)	744 MB	4,747,963	Supercomputer
HDFS	1.58 GB	11,175,629	Distributed Cluster

Table 2. Anomaly Detection Accuracy on HUFLIT Campus Logs

Model	Precision	Recall	F1-Score	Latency (ms)
PCA	0.7420	0.6850	0.7123	0.0045
DeepLog (LSTM)	0.8850	0.9120	0.8983	0.0240
LogAnomaly	0.9010	0.9240	0.9123	0.0310
TCN-Transformer (Ours)	0.9480	0.9560	0.9520	0.0125

3.3 LLM RCA Narration Quality & Triage Efficiency

Evaluating LLM-generated RCA narratives against expert-annotated incident playbooks yields an average ROUGE-L score of 0.8420 and BLEU-4 of 0.7815. Automated narration reduces human incident triage time from 15.4 minutes to 1.15 seconds per incident.

4 Conclusion

This paper presented an LLM-Augmented AIOps framework for log anomaly detection and automated Root Cause Analysis on real-world enterprise campus infrastructure. By integrating Drain3 streaming parsing, TCN-Transformer sequence modeling, and lightweight quantized LLM narration, the system achieves sub-second explainable incident triage ($F1 = 0.9520$, RCA latency = 1.15 s) evaluated on 5GB production campus logs.

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